See First, Refuse When Thin: Compiling Point-in-Time World Bundles for LLM Agents in Crypto Markets

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Abstract

Purpose. In AI-based finance, the performance frontier is often set less by model complexity than by whether the market world fed to the model is broad, stable, and honest. We treat the missing piece as a *data end*: an anonymized observation runtime that compiles multi-band crypto evidence into a point-in-time (PIT) *world bundle* (BandPIT/WMI) with hard refusal when support is thin—infrastructure for LLM/agent market analysis and for evaluating whether models *understand* that world, not a trading engine. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. ≲0.23 under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: under a *frozen full-schema* stress gate (max WMI ≈ 0.093<0.2), Compiled thin-abstain is 1.0 (100-day and full OOS n=2000) while Ungated soft judgment is only ≈0.68 / ≈0.56; Raw over-trades. Secondary—handoff: the *scoped-archive production valve* opens on ≈61% OOS days; live LLMs abstain 1.0 on closed days and act on ≈89% of open days, while a no-LLM tilt rule remains nonempty (Sharpe 0.77). **Claim.** Move from model-first to data-end-first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies → Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

AI in finance, LLM agents, data end, observation layer, world bundle, grounding, market understanding, cryptocurrency, point-in-time, quality-gated refusal

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1 Introduction

Purpose (read this first). At the methodology–application interface of AI and finance, many LLM/agent failures are not "the model is not smart enough," but "the market worldview is too thin." The stack is **data/observation → decision → execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. Stress episodes with multi-hour exchange outages make the gap concrete: an agent that sees only lagged momentum can act while

Table 1: AI-finance stack; this paper supplies the data end for LLM/agent consumers.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

the observation set is incomplete. We build the missing substrate—compile a PIT world bundle so models can *see* the market, evaluate that seeing with checkable grounding, and hard-refuse when the view is too thin. The product is reduced world-model error and auditable AI judgments, not trading answers.

Failure modes. No feed → refuse (safe, useless as a data end). Thin price fragment → trade and lose (unsafe for any downstream agent). Disclosed but ungated world → read fields, disagree on action. So the data end must (i) make the market legible to LLMs/agents and (ii) enforce a quality valve before decision/execution.

What we build. An anonymized *data-provision runtime for public LLMs and finance agents* (Section 4): multi-band evidence → BandPIT/WMI → JSON world bundle for GPT /DeepSeek /GLM /Gemini-class consumers, served through a read API. It is also an evaluation substrate: the same bundle supports grounding and gated-action protocols that test whether a model understands disclosed market state. "World model" means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; the hard abstain flag is the machine-readable valve.

How we evaluate (dual protocol). **RQ1 (primary—seeing):** Are ready /missing /tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Under a *frozen full-schema stress gate* (never clears 0.2), does hard policy enforce thin abstention where soft MAY does not? Arms: Compiled, HPO (hard prompt on numeric WMI, no boolean), Ungated, Raw, Blind. **RQ3a (handoff):** Under the *scoped-archive production valve*, do live LLMs obey closed days and act when open? Does a transparent no-LLM rule remain callable? **RQ3b (content):** On the same vintaged tilts, does an ungated content probe beat momentum (ΔCE; LOBO)?

Contributions. (1) AI–finance data-end semantics: epistemic observations, compilation Π_t, world-quality index with band scope, and seeing vs. gating—interfaces that let LLM/agent stacks consume crypto markets safely. (2) An anonymized data-provision path for agents: collectors to BandPIT to service surface (bundle / health / handoff) to world bundles, plus a frozen four-arm harness to *evaluate* market understanding. (3) Live LLM evidence that the substrate

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works: grounding ≈0.97; stress-gate abstain 1.0; scoped production valve obedience (closed 1.0; open act ≈0.89); nonempty tilt handoff (Sharpe 0.77; LOBO 1.0). Estimand: whether public models can see a typed market world, refuse when thin, and hand off when ready—not portfolio alpha.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance (ICAIF intersection). Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. Methodologically they still assume a usable observation stream; application stacks then conflate bad market context with bad policy. We target that gap in a data-centric sense: compile a typed PIT world for LLM/agent consumers, and score market understanding (grounding, evidence-bound refusal) before debating alpha. The estimand is orthogonal to memory routers and debate judges: without a typed world, those systems evaluate agents on tool fragments, not on whether the market view was broad, stable, and honest.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to *world* quality $q(W_t)$ via a runtime field, not classifier confidence alone (Prop. 3.9)—blocking the overconfident-when-thin failure mode. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness (Prop. 3.7). Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation-Layer Semantics

This section states the data-end theory as *interface semantics*: what counts as an observation, how compilation yields a world, and when quality forces refusal. It is not a pricing theorem and not a generative dynamics model. Systems realization is Section 4; empirical tests are Section 5.

Definition 3.1 (Epistemic observation). An observation is the tuple $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ (value, latest-available time, quality, main-view gate, semantic role). Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views into a consumer object.

Definition 3.3 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ with scalar quality $q(W_t)$; $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$. W_t is the handoff object to decision.

Definition 3.4 (World-quality index). Ready, limited, and missing counts disclose coverage. $\mathrm{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty in $[0, 1]$; ACWMI multiplies a regime weight. Band scope selects the support of (B, U, H): full-schema stress audit vs. the declared consumer archive (production valve). Scoped WMI scores only declared bands; it is a contract-completeness index, not a lowered $q^\star$. The Compiled interface exposes a hard abstain boolean $\mathbf{1}\{q(W_t) < q^\star\}$ (and, under archive scope, incompleteness) with a thin-world label (production $q^\star = 0.2$ for scoped WMI).

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t) < q^\star$. Tilts without a gate are a dump; a gate without a readable bundle is only a switch. The two interfaces are separable: Ungated keeps W_t and numeric $q(W_t)$ but removes the hard boolean. This is not feature engineering: a feature vector asks the model to infer reliability; a world bundle *states* clocks, missingness, and a machine-readable valve.

Assumption 3.5 (Bounded lag and increments). Band j has max observation lag Δ_j; latent increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Assumption 3.6 (Multiplicative quality independence). World availability requires breadth, stability, and honesty *jointly*: if any factor is 0, then $q(W_t) = 0$. The product WMI $= BUH$ is the simplest continuous $[0, 1]$ score obeying this axiom; averages can stay large when one factor vanishes.

Proposition 3.7 (Lag–missingness reconstruction bound). *Under Assumption 3.5, any compiler that uses only in-clock observations satisfies*

$$\|\widetilde{S}_t - S_t\| \;\le\; C_{\mathrm{del}}\bar{\delta}\sum_j \Delta_j + C_{\mathrm{noi}}\varepsilon_t + C_{\mathrm{miss}}m_t,$$

where m_t is disclosed missing mass and ε_t collects measurement noise. Operationally, BandPIT maps age to ready/limited/missing so m_t and lag terms are readable in W_t; compilation cannot erase delay, but it can refuse to pretend the bound is zero.

Proposition 3.8 (Compilation ≠ dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\mathrm{AI}}$: ungated stale evidence can raise volume while lowering honesty, tightening rather than loosening Prop. 3.7.*

Proposition 3.9 (World-conditional abstain). *Let $c_{\mathrm{act}}(W_t)$ be the expected cost of acting and c_{abs} the abstain cost. If $c_{\mathrm{act}}(W_t) > c_{\mathrm{abs}}$ and c_{act} is nonincreasing in $q(W_t)$, the Bayes action is abstain on $\{W : q(W) < q^\star\}$ [4]. A hard runtime contract implements that set; soft "you may abstain" text does not.*

Remark 3.10 (Threshold $q^\star = 0.2$). *Production valve: archive complete and* WMI $\ge q^\star$. *For $q^\star \in [0.10, 0.30]$ open share stays ≈0.61 (thick days ≈ 1; gap days fail completeness; gap median ≈0.14). A WMI-only counterfactual moves when $q^\star$ crosses that gap mass. We keep* 0.2 *above gap-day WMI—also where Ungated soft thin-abstain fails (≈0.68; Sec. 5.3).*

Proposition 3.11 (LOBO content vs. gating). *Deleting band j changes a downstream value functional through a content channel and a gating channel. Under an ungated rule the gating channel is shut, so the value gap identifies nonempty fields.*

Theory ⟶ protocol. Prop. 3.7–3.8 ⇒ RQ1. Prop. 3.9 ⇒ RQ2 (Compiled/HPO hard policy vs. Ungated soft MAY). Def. 3.4+Rem. 3.10 ⇒ RQ3a (scoped valve). Prop. 3.11 ⇒ RQ3b.

Remark 3.12 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface: the data end of the stack.*

4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path for LLM/agent consumers (Figure 1, Table 2). Decision and execution sit downstream; this paper supplies their market context and the protocols that test whether that context was understood. Paths and product names are omitted for double-blind review.

4.1 Layered design

(1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
(2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
(3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty and missingness gates implement Π_t; WMI, ACWMI, and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle reads, a valve-gated handoff endpoint (tilt action, no LLM), health exposing the abstain flag with band scope, a paper world-model time slice, and availability-shock queries.
(5) **Consumer /eval layer.** Frozen-prompt public LLMs (and rule agents) call an OpenAI-compatible adapter at temperature 0 under four arms—the harness that scores market understanding against the supplied world.

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \longrightarrow$ WMI; band scope; abstain; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / handoff / availability reads
Consumer harness	Four-arm protocol; temp. 0

4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of ≈3990 asset-days (2025-07 to 2026-08): exchange ready share ≈0.80 panel-wide (≈0.61 OOS), macro 1.0, alternative 1.0. **Full-schema stress WMI** scores permanently empty non-archive slots and caps at max ≈0.093, so a 0.2 gate never opens—the frozen LLM-arm protocol (RQ2). **Scoped-archive production WMI** renormalizes B, U, H to {exchange, macro, alternative} and opens iff the archive is complete and WMI $\geq$ 0.2 (RQ3a); this measures contract readiness, not a relaxed threshold.

4.3 PIT clock and freshness

For calendar day t, BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the bundle records `decision_asof`. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.7). Breadth, stability, and honesty aggregate into $\text{WMI}_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The *scoped production valve* sets `should_ai_abstain` when the archive is incomplete or scoped WMI < 0.2, and exposes `thin_world`; bundles also disclose `full_schema_wmi` as a ceiling audit. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: same panel, two scopes—full-schema stress WMI stays under 0.2; scoped production WMI clears 0.2 on archive-complete days (shaded). Table 5: scoped production valve opens on ≈61% OOS days (≡ 3/3 ready); full-schema stress gate stays closed.

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t, evidence IDs

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

Table 5: Dual gates vs. open share (OOS). Production = scoped; stress = full schema.

Gate	Open share	Note
Scoped production 0.2	0.61	complete ∧ WMI≥ 0.2
Full-schema stress 0.2	0.00	max WMI ≈0.093

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and `evidence_ids`. Listing 1: exchange-gap day (scoped valve closed). Listing 2: 3/3-ready day where scoped WMI=1.0 opens the production valve; `full_schema_wmi`≈0.093 is disclosed as ceiling audit—same day, two scopes, not a lowered $q^{\star}$.

Listing 1: Exchange-gap bundle (Compiled; scoped closed).


```
{"date":"2026-05-11","asset":"ADA",
```

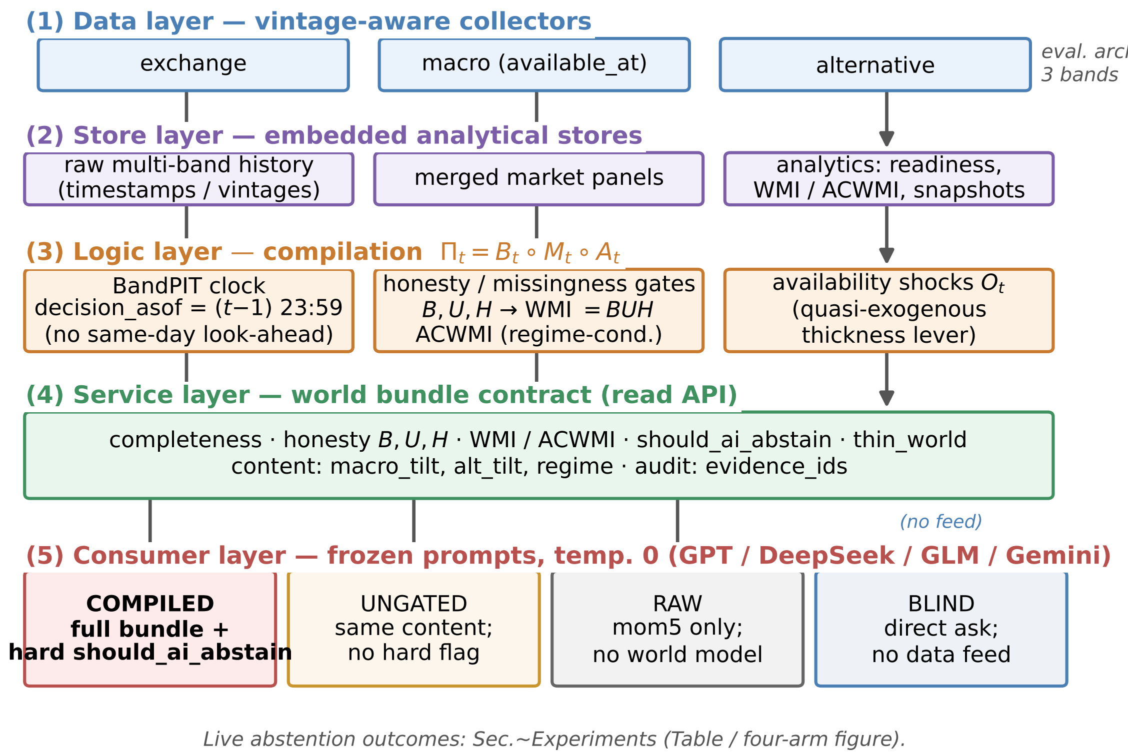

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

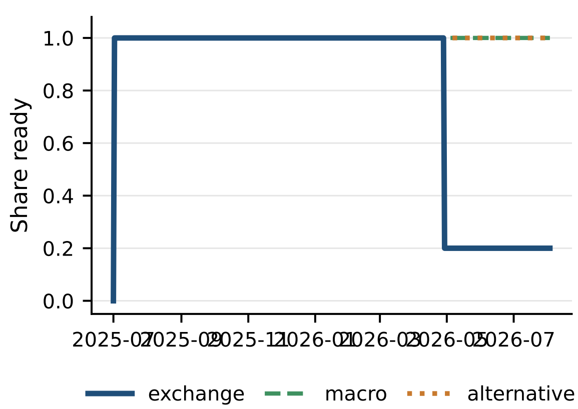

Figure 2: PIT readiness: what is visible in the archive over time.


```
"decision_asof":"2026-05-10T23:59:00",
"completeness":{"n_ready":2,"n_missing":1},
"world_model_index":{"band_scope":"eval_archive",
  "wmi":0.139,"archive_complete":false,
  "should_ai_abstain":true,"thin_world":true,
  "full_schema_wmi":0.015},
"macro_tilt":1.0,"alt_tilt":-1.0}
```


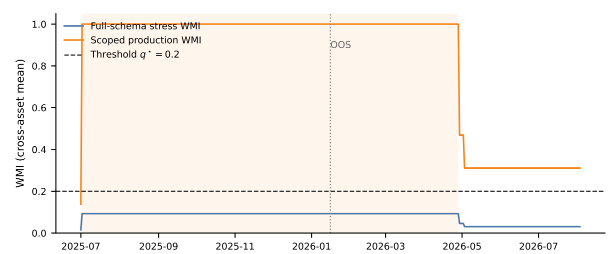

Figure 3: Dual-scope WMI on the same panel: stress ceiling vs. production contract (shaded = archive-complete).

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H; main-view / stale gates
Quality	WMI, ACWMI, band scope, abstain, thin-world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence IDs; EAR; full-schema WMI

Listing 2: 3/3-ready day: scoped valve opens.


```
{"date":"2026-01-16","asset":"ADA",
```

```
"completeness":{"n_ready":3,"n_missing":0},
"world_model_index":{"band_scope":"eval_archive",
  "wmi":1.0,"archive_complete":true,
  "should_ai_abstain":false,"thin_world":false,
  "full_schema_wmi":0.093},
"band_status":{"exchange":"ready","macro":"ready",
  "alternative":"ready"},
"macro_tilt":1.0,"alt_tilt":-1.0}
```


4.5 Arms and frozen prompts

Actions ∈ {bullish, bearish, neutral, abstain}. **Compiled:** full bundle; prompt *requires* abstain when the hard boolean is true. **HPO** (hard-prompt-only): same numeric fields as Ungated (*no* boolean); prompt *requires* abstain if WMI < 0.2. **Ungated:** same content and numeric WMI; soft prompt (*may* abstain). **Raw:** mom5 only. **Blind:** date+asset, no feed. **Dual protocol.** RQ1–RQ2 freeze full-schema bundles (max WMI ≈ 0.093); RQ3a switches to scoped production bundles. Prompts, IS/OOS cut, and seeds are frozen (Listings 3–4).

Listing 3: Compiled (excerpt): hard boolean.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum.
```

Listing 4: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI makes action unsafe;
   you MAY "abstain".
NO hard boolean; NO forced policy.
```

Why the hard valve is not a confound. The data end must *block* thin-day action (Prop. 3.9). HPO separates hard *prompt* obedience on numeric WMI from the typed boolean; Ungated removes both; Raw/Blind bound the ladder.

4.6 Systems validation (engines)

Paper-lab smoke wires BandPIT/WMI/ACWMI and checks the dual gate: scoped archive opens on 3/3-ready statuses while full-schema stays closed; a handoff rule acts only when the scoped valve is open. Availability shocks O_t are queryable on the service surface.

5 Experiments

5.1 Setup

PIT panel ~399 days × 10 liquid names (≈3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses n=2000; no-LLM probes use ≈200 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain under the *full-schema stress gate* (CE as avoided loss, not alpha); RQ3a scoped production-valve LLM obedience plus no-LLM tilt handoff; RQ3b ungated tilt ΔCE vs. momentum.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

Model	Ready F1 C	Ready F1 R	Missing F1 C	Missing F1 R	Tilt-sign C	Tilt-sign R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility ≠ sufficiency chat. Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers "enough?" correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of "is this enough to trade?" is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) under the *full-schema stress* Compiled arm abstains on `should_ai_abstain`; Raw goes bullish from mom5 = +1 alone.

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

Table 9: Thin abstention on identical 100 OOS days under full-schema stress. Blind = 1.0; HPO = hard prompt on numeric WMI, no boolean.

Model	C	HPO	U	Abs. R	ΔCE
gpt-5.4-mini	1.00	1.00	0.63	0.04	+0.73
deepseek-v4-flash	1.00	1.00	0.81	0.75	+1.22
glm-5.2	1.00	1.00	0.86	0.11	+1.04
gemini-3.5-flash-lite	1.00	1.00	0.43	0.07	+1.30
Mean	1.00	1.00	0.68	—	+1.07

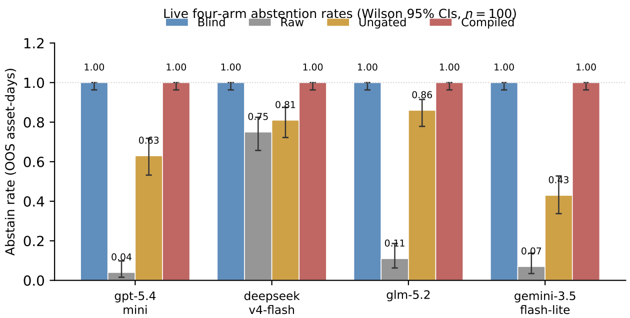

Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled/HPO enforce hard policy.

Table 10: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. All live LLM arms here freeze *full-schema* bundles, so every OOS day is thin under the stress gate 0.2. Table 9 and Figure 4: on the same 100 days, Compiled and HPO abstain 1.0; Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. HPO shows hard *numeric* prompting recovers perfect thin abstention when WMI is unambiguously below 0.2; Ungated shows soft MAY does not. The typed boolean remains the API/non-LLM contract; the ladder shows that *hard policy*, not soft disclosure, is what binds. Offline soft evidence-alignment (rationale cues to disclosed bands/tilts/quality) averages ≈0.998 Compiled / 0.993 HPO vs. 0.80 Ungated and 0.38 Raw on matched 100-day slices. Mean ΔCE (Compiled−Raw) +1.07 is avoided loss, not alpha.

Full-OOS replication. Table 11: on all 2000 OOS asset-days (all thin under the *full-schema stress* gate 0.2), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈0.56; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor): the failure mode that needs a data end is not "models never abstain," but "thin fragments make them act."

Table 11: Full-OOS thin abstention (n=2000; all thin under full-schema stress gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈0.56	—

*GLM coverage incomplete (Ungated n=1402; Raw n=675).

Table 12: Open slice under full-schema LLM arms (3/3 ready; n≈1224). Soft-judgment lower bound.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM n=398; Ungated GLM n=932; others n=1224. Bootstrap mean CE ≈+0.31 (B=5, 999 reps; one-sided p≈0.21).

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek −3.12, GLM −4.28, Gemini −1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

Open slice (still under full-schema LLM arms). Table 12: on the band-thick subset (3/3 ready ≡ full-schema WMI ≥ 0.05; ≈61% OOS; n≈1224) Ungated keeps compiled content without the hard boolean—mean abs. ≈0.48, CE positive for every vendor (mean ≈+0.29; bootstrap ≈+0.31, CIs include 0). Full-schema Compiled still abstains at 1.0 here; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). The same thick days are where the *scoped production valve* opens; live scoped LLM handoff is RQ3a.

5.4 RQ3a (handoff): Scoped production valve with live LLMs

Having established seeing (RQ1) and stress-gate refusal (RQ2), we ask whether the *production* valve is callable. The full-schema stress gate never clears 0.2, so it cannot open—by design in RQ2. Scoped production WMI opens on ≈61% OOS days (≡ 3/3 ready). We re-run the same four vendors on a stratified sample of 70 open + 30 closed OOS asset-days with scoped bundles (seed fixed; temp. 0).

Table 13: on closed days Compiled abstains 1.0 for every vendor (Ungated soft rates ≈0.87–0.97; Raw often keeps trading). On open days Compiled acts on a mean ≈89% of days—the valve opens and models consume the bundle. Open-day CE does *not* beat Raw (mean

Table 13: RQ3a live LLMs under scoped production valve (70 open + 30 closed).

Model	Abs C_{cl}	Abs C_{op}	CE C_{op}	CE R_{op}	ΔCE_{cl}
gpt-5.4-mini	1.00	0.03	−4.00	+0.09	+0.49
deepseek-v4-flash	1.00	0.24	−5.22	−0.21	+0.17
glm-5.2	1.00	0.11	−4.31	−0.39	+3.06
gemini-3.5-flash-lite	1.00	0.07	−4.27	−0.46	+2.64
Mean	1.00	0.11	–	–	+1.59

Abs $C_{cl/op}$: Compiled abstain on closed/open days. ΔCE_{cl} = avoided loss vs. Raw on closed days.

Table 14: RQ3a no-LLM handoff. Scoped production valve vs. full-schema stress.

Policy (reads bundle fields)	Sharpe	CE†	Abstain
Full-schema stress 0.2	0.000	0.000	1.00
Scoped production 0.2 + tilts	**0.772**	**+0.253**	0.39
Momentum (mom5)	0.101	−0.075	0.00
Buy-and-hold	−1.399	−0.987	0.00

†CE = ann. mean − $\frac{1}{2}$ ann. variance (valve-aware path).

Table 15: RQ3b ungated content probe (no valve). Bootstrap ΔCE 0.334.

Policy	Sharpe	CE‡
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	**0.764**	**+0.130**
Tilts − Momentum	–	0.334 (p=0.034)

‡Content-harness CE (always-on; not comparable 1:1 to †).

ΔCE≈−4.2): opening the data-end valve is not an LLM alpha claim; policy skill remains downstream. Closed-day refusal avoids Raw losses (mean ΔCE≈+1.6 vs. Raw on the closed slice).

A transparent no-LLM rule on the same vintaged tilts remains the clean nonempty-field handoff (Table 14): Sharpe 0.772 / CE +0.253 when the scoped valve is applied, versus momentum and buy-and-hold; full-schema stress stays fully closed.

5.5 RQ3b (content): Nonempty fields

Table 15 and Figure 5: ungated always-on tilts vs. momentum on the same vintaged fields (ΔCE=0.334, p=0.034; LOBO content share 1.0). This probe shuts the gating channel (Prop. 3.11); it is not the production valve.

Density split and LOBO channel. Table 16: open/thick ΔCE=0.315; on exchange-gap days the always-on probe still trades (high slice CE is a selection artifact, not a valve claim; ΔCE=0.129). Table 17 and Figure 6 (Prop. 3.11): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).

6 Discussion

What carries the paper. (i) The compiled world is legible (grounding ≈0.97; sufficiency chat is not a substitute). (ii) Hard policy binds on thin days (Compiled/HPO 1.0); soft MAY does not (≈0.68). (iii) Scoped production opens ≈61% OOS days; LLMs obey the valve; open-day CE ≠ alpha; no-LLM tilts stay nonempty. Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

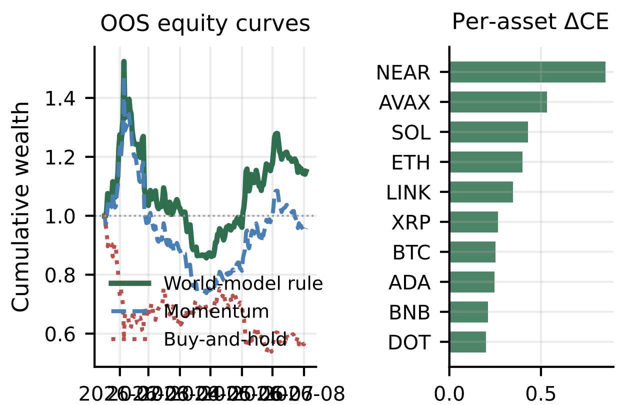

Figure 5: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.

Table 16: RQ3b content probe by archive density (OOS; ungated).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048§	0.129

*Block-bootstrap ΔCE; raw mean gap 0.336. §Always-on selection artifact.

Table 17: LOBO decomposition (ungated content probe). Content share 1.0.

Drop	ΔCE content	p	Content share
macro	−0.334	0.034	1.0
alternative	−0.334	0.034	1.0
exchange	0.000	1.000	–

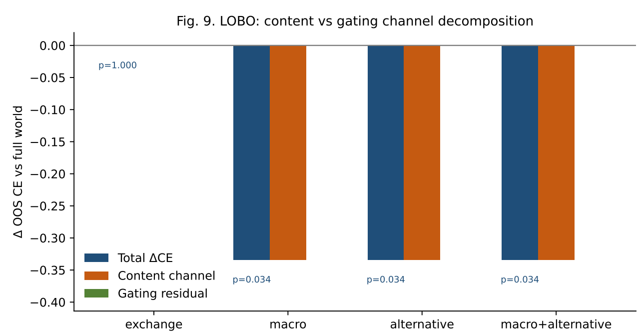

Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.

Deployment path. Level 1: observation API (PIT world bundles; this paper). Level 2: community benchmark for market understanding (grounding, gated action, soft EAR). Level 3: compliance logging of abstain decisions for algorithmic accountability. We do not claim to solve crypto prediction or open-day LLM alpha; seeing a typed world reduces decision risk before models get "smarter."

Design and scope. World bundle ≠ feature dump; previous-close PIT; dual WMI scope; HPO separates hard prompt from hard boolean. Open-day LLM CE is a usability check—policy skill sits downstream of the data end. Prompts, cut, and seeds are frozen; an anonymized harness ships on acceptance.

7 Conclusion

AI-in-finance LLM/agent stacks need a reliable data end before decision or execution. We formalize that observation layer and show: grounding $\approx$0.97; hard policy (Compiled/HPO) thin-abstains 1.0 while soft Ungated does not; scoped production opens and is obeyed; tilts remain nonempty without claiming open-day LLM alpha. Move from model-first to data-end-first: compile so models can see; refuse when they must not act.

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